| Expt. 1<br>22/6/26 | Study of various Network commands used in Linux and Windows |
|---|---|

Basic Networking commands:-

1) arp -a:- ARP is short form of address resolution protocol, It will show the IP address of your computer along with the IP address and MAC address of your router.
   - Ex:- Gateway (10.0.2.2) at 52:54:00:12:35
2) hostname:- This is the simplest of all TCP/IP commands. It simply displays the name of your computer.
   output: localhost.localhost.localdomain
3) ipconfig/all: This command displays detailed configuration information about your TCP/IP connection including router, Gateway, DNS, DHCP, and type of ethernet adapter in your system.
   output: Windows IP configuration
   Ethernet adapter Ethernet 2:
   connection-specific DNS suffix :
   link-local IPv6 Address ... : fe80::6aa0:73f8:648d:a46e%3
   Subnet Mask ..... : 192.168.56.1
   Default Gateway .....
4) nbtstat -a: This command helps solve problems with NetBIOS name resolution.
   output: Displays protocol statistics and current TCP/IP connections using NBT.
   NBTSTAT [[-a Remote Name] [-A IP address] [-c] [-n] [-r] [-R] [-RR] [-s] [-S] [interval]]

5) netstat: (network statistics) netstat displays a variety of statistics about a computers active TCP/IP connecting. It is a command line tool for monitoring network connections both incoming and outgoing as well as viewing statistics etc.

output:-

Active connections:

| Proto | Local Address | Foreign Address | State |
|---|---|---|---|
| TCP | 10.55.47.189:50357 | 4.213.25.240:https | Established |
| TCP | 10.55.47.189:50358 | 4.213.25.240:https | Established |

6) nslookup: It is a tool used to perform DNS lookups in Linux. It is used to display DNS details, such as the IP address of a particular computer the MX record for a domain.

output: Default Server: unknown
Address: 10.55.47.246.

7) Pathping: Pathping is unique to Windows and is basically a combination of the ping and Tracert commands.

output: pathping [-g host-list] [-h maximum_hops]
[-i address] [-n] [-P Period] [-q num-queries]
[-w timeout] [-4] [-6] target_name.

) Ping: (Packet INternet Groper) command is the best way to test connectivity between two nodes.

output: usage: Ping [-t] [-a] [-n count] [-l size] [-f]
[-V TOS] [-r count] [-s count] [-j host-list]

9) Route : Route command is used to show / manipulate the IP routing table.

output:

Kernel IP routing table

| Destination | Gateway | Genmask | Flags | Metric | Ref |
|---|---|---|---|---|---|
| default | - Gateway | 0.0.0.0 | UG1 | 100 | 0 |
| 10.0.2.0 | 0.0.0.0 | 255.255.255.255 | U | 100 | 0 |

Some important linux networking commands

1. ip : The ip command can show address information, manipulate routing, plus display network various devices, interfaces and tunnels.

~~output :~~ i) ip address show

output :

1: lo: <LOOPBACK, UP, LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000

ii) To assign an IP to an interface for example enp503

2. ifconfig

The ifconfig command was/is a staple in many sysadmin's tool, belt for configuring and troubleshooting between network.

output:- flags = 4163 <UP, LOOPBACK, RUNNING, MULTICAST>
mtu 1500 inet init 10.0.2.15 netmask 255.255.2255.0

3) mtr

MTR is a program with a command line interface that serves as a network diagnostic and troubleshooting tools. This command combines the functionality of the ping and troubleshooting tool.

output:-

My traceroute [v0.94]

Host

1.localhost Packets Pings.

| | Loss% | Snt | Last | Avg | Best | Wrst | StDev. |
|---|---|---|---|---|---|---|---|
| | 0.0% | 122 | 0.1 | 0.2 | 0.1 | 1.9 | 0.2. |

a. # mtr google.com

O/P:- host

1. gateway
2. (waiting for reply)

b. # mtr:-g google.com

O/P:-

usage:

mtr [options] hostname

-F, --filename FILE read hostname(s) from a file.

) tcpdump

The tcpdump command is designed for capturing and displaying packets.

O/P:- tcpdump: ~~ens~~ ens POS3: you don't have permission to capture on that device.

5) Ping : Ping is a tool that verifies IP-level connectivity to another TCP/IP computer by sending internet control message protocol (ICMP) Echo request messages.

i/p: Ping google.com

o/p:- PING google.com (142.250.182.78) 56(84) bytes of data
64 bytes from lcmaaa-ax-in-f14.1e100.net (142.250.182.78):
icmp_seq=1. ttl=112 time=938 ms.

Result:

Thus the various network commands used in linux and Windows was executed successfully.